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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/743,953	06/14/2024	Yu Fang	SONA-0012-U01-C10	7752

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EXAMINER	
ALGIBHAH, HAMZA N	

ART UNIT	PAPER NUMBER
2441	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@gtclawgroup.com

Office Action Summary	Application No. 18/743,953	Applicant(s) Fang et al.	
	Examiner HAMZA N ALGIBHAH	Art Unit 2441	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 06/14/2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-20 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 06/14/2024 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Detailed Action

1. ***Claims*** 1-20 are pending.

2. ***Claims*** 1-20 ***are rejected.***

Claim Rejections - 35 USC § 102

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale, or otherwise available to the public before the effective filing date of the claimed invention.

3. ***Claim***(s) 1-20 is/are rejected under 35 U.S.C. 102(a)(1) as being anticipated by Breed (Pub. No.: US 2014/0276090 A1).

As per claim 1, Breed discloses an apparatus, comprising:

- ***a controller*** (Breed, Fig 4 item 79, paragraph 0105, wherein an electronic control unit (ECU) 9 which performs the health monitoring calculations as explained below. ECU 9 includes software and hardware known to those skilled in the art, such as a processor and signal processing circuitry and software embodied on non-transitory computer-readable media to implement the processing stages described herein)***, comprising:***

- ***an automation definition circuit structured to interpret an automation user interface*** (Breed, paragraph 0099, wherein monitoring systems are imprecise and thus subject to false alarms and consequently, before any action is taken to take control of any vehicle systems, the diagnosis should ideally be verified through driver feedback. This feedback can take many forms such as employing a heads-up display on the windshield or elsewhere in the view of the driver that display a query asking the driver to respond, such as orally or by depressing the horn or

other switch or button, or making a gesture, for example. Other feedback systems include an oral alarm or verbal statement requesting a response from the driver or any other visual or audible alarm or message requiring the driver's attention and subsequent action indicating that the sensed event was a false alarm. Wherein, the user interface can be the interface displaying to the deriver the alarms or questions and/or receiving the driver feedback), ***and to interpret an automation request in response to user communications with the automation user interface*** (Breed, paragraph 0099, wherein monitoring systems are imprecise and thus subject to false alarms and consequently, before any action is taken to take control of any vehicle systems, the diagnosis should ideally be verified through driver feedback. This feedback can take many forms such as employing a heads-up display on the windshield or elsewhere in the view of the driver that display a query asking the driver to respond, such as orally or by depressing the horn or other switch or button, or making a gesture, for example. Other feedback systems include an oral alarm or verbal statement requesting a response from the driver or any other visual or audible alarm or message requiring the driver's attention and subsequent action indicating that the sensed event was a false alarm. Failing to receive a proper response, the driver monitoring system can repeat the request a number of times, wherein the specific number may be dependent on or determined by the urgency of the situation (i.e., a more urgent situation might require more requests than a less urgent situation). In one example, the driver monitoring system can ask the driver "are you alright?" and if the driver responds with something like "yes", then the system can be reset)

- ***an automation management circuit structured to determine an automation description in response to the automation request*** (Breed, Fig 9, paragraph 0136, wherein steps 151, 152 and 153 determine whether any of the measured heartbeat, respiration and motion rates are normal and if not, i.e., if one is not normal, a warning and feedback system is invoked

or activated at 154. If the driver fails a feedback test generated by the warning and feedback system, step 155, then a routine to control the vehicle is invoked or activated at 156)***; and***

- ***an automation command circuit structured to provide the automation description to a controller of a vehicle, wherein the controller of the vehicle is configured to implement an automated vehicle response, responsive to the automation description*** (Breed, Fig 9, paragraph 0136-0137, wherein the vehicle control routine 156 may also be performed partly or entirely by equipment on the vehicle upon command by a software program that is executed when needed. The equipment necessary for the control unit to invoke control of the vehicle includes, but is not limited to, devices that are coupled to the steering wheel or steering system, the brakes or braking system, the throttle system, transmission and/or engine. These devices may be controlled by software to reduce the speed of the vehicle to a stop and direct the vehicle to a safe area, e.g., the side or shoulder of a road. The vehicle control routine 156 may therefore correspond to or encompass a processor and control equipment that is coupled to vehicular parts that affect operation of the vehicle, e.g., the brakes, steering wheel and throttle, and possibly servos or actuators therefor)***; and***

- ***wherein the automation command is structured to implement an automated vehicle response via an end point of a plurality of end points of a network of a vehicle*** (Breed, Fig 9, paragraph 0136-0137, wherein the vehicle control routine 156 may also be performed partly or entirely by equipment on the vehicle upon command by a software program that is executed when needed. The equipment necessary for the control unit to invoke control of the vehicle includes, but is not limited to, devices that are coupled to the steering wheel or steering system, the brakes or braking system, the throttle system, transmission and/or engine. These devices may be controlled by software to reduce the speed of the vehicle to a stop and direct the vehicle to a safe area, e.g., the side or shoulder of a road. The vehicle control routine 156 may

therefore correspond to or encompass a processor and control equipment that is coupled to vehicular parts that affect operation of the vehicle, e.g., the brakes, steering wheel and throttle, and possibly servos or actuators therefor. Breed, Fig 3-4, paragraph 0113, 0117, wherein perspective view with parts cutaway and removed of a vehicle showing the passenger compartment containing electric field occupant sensing systems associated with both the driver and passenger seats is illustrated in FIG. 3. In addition, the plurality of components shown in Fig. 4 can be the plurality of end points as claimed}.

Examiner Note: The components labeled as the 'automation definition circuit,' 'automation management circuit,' and 'automation command circuit' are asserted to form integral parts of a unified controller. Breed 's system, as depicted in paragraph 0105 as an example, accomplishes the described functions through an 'n electronic control unit (ECU) ' Consequently, it is to be inferred that the ECU inherently encompasses all essential circuits required to execute the aforementioned functions.

As pre claim 2, claim 1 is incorporated and Breed further discloses **wherein the automation management circuit is further structured to confirm an authorization of a user for the automation request, and to determine the automation description in response to the authorization** (Breed, Fig 9, paragraph 0132, 0140, 0232, wherein the routine is illustrated generally at 180. A first step in the routine 180, at 181, is to initiate a warning set counter. The second step 182 is to generate and send a message in some form to get the attention of the driver. This message can be in any of several forms, e.g., audible, visual, tactile, such as a warning buzzer or a flashing light which can suffice, but this may confuse a driver that is unfamiliar with the system. If the vehicle is equipped with a heads-up display or other display that will get the attention of an alert driver, then a message can be displayed on one or more of such displays. A preferred message is to have a synthesized or recorded voice announce a

message to the driver. A combination of the above warnings can be better still. The message can instruct the driver to take some deliberate action which will not interfere with his ability to operate the vehicle but will nevertheless indicate that he has received the message and responded in a timely manner);

As pre claim 3, claim 2 is incorporated and Breed discloses **wherein the automation management circuit is further structured to confirm the authorization in response to at least one of: data to be collected for the automated vehicle response; data to be provided on a network of the vehicle for the automated vehicle response; an actuator to be accessed for the automated vehicle response; a service to be accessed for the automated vehicle response; a flow to be accessed for the automated vehicle response; an application to be accessed for the automated vehicle response; or an operational impact associated with the automated vehicle response** (Breed, Fig 9, paragraph 0134, 0136, wherein antenna routine 141 will look for the strongest antenna signals for heartbeat, respiration rate and occupant motion. Once selected, control passes to the modules that acquire respiration 142, acquire heartbeat 143, and acquire motion 144 and establish or record normal respiration 145, normal heartbeat 146, and normal motion frequency 147, what will be termed the established respiration rate, heartbeat rate and typical occupant motion. Steps 151, 152 and 153 determine whether any of the measured heartbeat, respiration and motion rates are normal and if not, i.e., if one is not normal, a warning and feedback system is invoked or activated at 154. If the driver fails a feedback test generated by the warning and feedback system,);

As pre claim 4, claim 2 is incorporated and Breed discloses **wherein the automation management circuit is further structured to determine the automation description as an**

amended version of the automation request in response to the authorization (Breed, paragraph 0099, 0141, wherein in one example, the driver monitoring system can ask the driver "are you alright?" and if the driver responds with something like "yes", then the system can be reset. If the driver fails to respond to a simple verbal request for a simple verbal response, the request can be changed to a more complex request such as "if you are alright, depress the horn button or raise your hand."; wherein the amended version of the automation request can be the request changed to a more complex request);

As pre claim 5, claim 4 is incorporated and Breed discloses **wherein the automation definition circuit is further structured to provide an amendment communication to the automation user interface in response to the amended version of the automation request** (Breed, paragraph 0099, 0141, wherein if the driver fails to respond to a simple verbal request for a simple verbal response, the request can be changed to a more complex request such as "if you are alright, depress the horn button or raise your hand." If the driver fails to respond appropriately to this more complex request, one or more of a variety of actions can be taken by the vehicle system depending on an assessment on the vehicle as to what is most appropriate);

As pre claim 6, claim 2 is incorporated and Breed discloses **wherein the automation management circuit is further structured to refuse the automation request in response to the authorization** (Breed, paragraph 0099, wherein in one example, the driver monitoring system can ask the driver "are you alright?" and if the driver responds with something like "yes", then the system can be reset. Wherein the yes response will reset the system which is a cancelation/refusing for the request for automation control of vehicle);

As pre claim 7, claim 6 is incorporated and Breed discloses **wherein the automation management circuit is further structured to provide a refusal communication to the automation user interface in response to the refusal of the automation request** (Breed, paragraph 0099, wherein if the driver fails to respond to a simple verbal request for a simple verbal response, the request can be changed to a more complex request such as "if you are alright, depress the horn button or raise your hand." Wherein the pressing of the horn and or raising of the hand can be the refusal communication that is provided to the user interface);

As pre claim 8, claim 1 is incorporated and Breed discloses **wherein the automation command circuit is further structured to provide the automation description as one of a policy or a policy update** (Breed, paragraph 0070, 0149, wherein a pattern of the blood vessel, capillary and vein in the face of the occupant may be monitored to provide a biometric identifier of the driver, and then adjustment of a vehicular system, such as a mirror, seat, seatbelt, radio, etc., controlled based on the biometric identifier of the driver. The processor would perform this analysis to identify the driver and then, control access to the vehicle, adjust seating conditions, adjust environmental conditions, etc., based on pre-set information about the biometrically-identified driver, wherein the pre-set information about the biometrically-identified driver can be the policy);

As pre claim 9, claim 1 is incorporated and Breed discloses **wherein the automation command circuit is further structured to receive a confirmation communication from the controller of the vehicle** (Breed, paragraph 0093, 0135 wherein a system that can work with eye trackers or independently is thus needed to sense cases missed by eye trackers or to confirm the

eye tracker results in order to minimize false alarms. Establishment of a normal motion is the least important of the three since it will mainly become important when an anomaly has been detected in the heartbeat or respiration rates. If the driver has stopped his normal motion when there is a significant heartbeat event, then this may confirm that there may be a problem with the driver, rather than a failure of the electric field monitoring system 121. Also, significant motion may be a clue that the select antenna routine 141 requires reactivation. Thereafter, the respiration, heartbeat and motion are acquired at 148, 149, 150, respectively);

As pre claim 10, claim 1 is incorporated and Breed discloses **wherein the automation definition circuit is further structured to provide a confirmation communication to the automation user interface** (Breed, paragraph 0124, 0140-0141, wherein health monitoring system ECU 91 can also receive information from the electric field monitoring system ECU 9, via a coupling or electrical connection therebetween (not shown), and notify the driver that the electric field monitoring system detects that he or she may no longer be capable of operating the vehicle and ask for feedback. A message to the driver can be in any form, such as a display, sound, light or any other system that can get the attention of the driver. The second step 182 is to generate and send a message in some form to get the attention of the driver. This message can be in any of several forms, e.g., audible, visual, tactile, such as a warning buzzer or a flashing light which can suffice, but this may confuse a driver that is unfamiliar with the system. If the vehicle is equipped with a heads-up display or other display that will get the attention of an alert driver, then a message can be displayed on one or more of such displays. A preferred message is to have a synthesized or recorded voice announce a message to the driver. A combination of the above warnings can be better still. The message can instruct the driver to take some deliberate

action which will not interfere with his ability to operate the vehicle but will nevertheless indicate that he has received the message and responded in a timely manner);

Claims 11-20 are rejected under the same rationale as claims 1-10.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to HAMZA N ALGIBHAH whose telephone number is (571)270-7212. The examiner can normally be reached 7:30 am - 3:30 pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Wing Chan can be reached on (571) 272-7493. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or

571-272-1000.

/HAMZA N ALGIBHAH/
Primary Examiner, Art Unit 2441